

Package ‘PReMiuM’

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Type Package

Title Dirichlet Process Bayesian Clustering, Profile Regression

Version 3.2.14

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Description

Bayesian clustering using a Dirichlet process mixture model. This model is an alternative to regression models, non-parametrically linking a response vector to covariate data through cluster membership. The package allows Bernoulli, Binomial, Poisson, Normal, survival and categorical response, as well as Normal and discrete covariates. It also allows for fixed effects in the response model, where a spatial CAR (conditional autoregressive) term can be also included. Additionally, predictions may be made for the response, and missing values for the covariates are handled. Several samplers and label switching moves are implemented along with diagnostic tools to assess convergence. A number of R functions for post-processing of the output are also provided. In addition to fitting mixtures, it may additionally be of interest to determine which covariates actively drive the mixture components. This is implemented in the package as variable selection. The main reference for the package is Liverani, Hastie, Azizi, Papathomas and Richardson (2015) <[doi:10.18637/jss.v064.i07](https://doi.org/10.18637/jss.v064.i07)>.

URL <https://www.silvialiverani.com/software/>

License GPL-2

LazyLoad yes

Depends R (>= 3.5.1)

Imports Rcpp (>= 0.12.13), ggplot2 (>= 2.2), cluster, plotrix (>= 3.6-6), data.table (>= 1.10.4-3), spdep (>= 0.7-7), sf (>= 1.0-8)

Suggests testthat (>= 1.0.2)

LinkingTo Rcpp, RcppEigen (>= 0.3.3.3.0), BH (>= 1.65.0-1)

SystemRequirements GNU make

NeedsCompilation yes

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PRemiuM-package	<i>Dirichlet Process Bayesian Clustering</i>
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Description

Dirichlet process Bayesian clustering and functions for the post-processing of its output.

Details

Package:	PRemiuM
Type:	Package
Version:	3.2.9
Date:	2023-06-02
License:	GPL2
LazyLoad:	yes

Program to implement Dirichlet Process Bayesian Clustering as described in Liverani et al. 2014. This is a package for Bayesian clustering using a Dirichlet process mixture model. This model is an alternative to regression models, non-parametrically linking a response vector to covariate data through cluster membership. The package allows Bernoulli, Binomial, Poisson, Normal, survival

and categorical response, as well as Normal and discrete covariates. It also allows for fixed effects in the response model, where a spatial CAR (conditional autoregressive) term can be also included. Additionally, predictions may be made for the response, and missing values for the covariates are handled. Several samplers and label switching moves are implemented along with diagnostic tools to assess convergence. A number of R functions for post-processing of the output are also provided. In addition to fitting mixtures, it may additionally be of interest to determine which covariates actively drive the mixture components. This is implemented in the package as variable selection.

The R package PReMiuM is supported through research grants. One key requirement of such funding applications is the ability to demonstrate the impact of the work we seek funding for can. Whatever you are using PReMiuM for, it would be very helpful for us to learn about our users, to tailor our future methodological developments to your needs. Please email us at liveranis@gmail.com or visit <http://www.silvialiverani.com/support-premium/>.

Details

PReMiuM provides the following:

- Implements an infinite Dirichlet process model
- Can do dependent or independent slice sampling (Kalli et al., 2011) or truncated Dirichlet process model (Ishwaran and James, 2001)
- Handles categorical or Normal covariates, or a mixture of them
- Handles Bernoulli, Binomial, Categorical, Poisson, survival or Normal responses
- Handles inclusion of fixed effects in the response model, including a spatial CAR (conditional autoregressive) term
- Handles Extra Variation in the response (for Bernoulli, Binomial and Poisson response only)
- Handles variable selection (tested in Discrete covariate case only)
- Includes label switching moves for better mixing
- Allows user to exclude the response from the model
- Allows user to compute the entropy of the allocation
- Allows user to run with a fixed alpha or update alpha (default)
- Allows users to run predictive scenarios (at C++ run time)
- Basic or Rao-Blackwellised predictions can be produced
- Handling of missing data
- C++ for model fitting
- Uses Eigen Linear Algebra Library and Boost C++
- Completely self contained (all library code is included in distribution)
- Adaptive MCMC where appropriate
- R package for generating simulation data and post processing
- R plotting functions allow user choice of what to order clusters by

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References

Molitor J, Papathomas M, Jerrett M and Richardson S. (2010) Bayesian Profile Regression with an Application to the National Survey of Children's Health, *Biostatistics* 11: 484-498.

Papathomas M, Molitor J, Richardson S. et al (2011) Examining the joint effect of multiple risk factors using exposure risk profiles: lung cancer in non smokers. *Environmental Health Perspectives* 119: 84-91.

Hastie, D. I., Liverani, S., Azizi, L., Richardson, S. and Stucker I. (2013) A semi-parametric approach to estimate risk functions associated with multi-dimensional exposure profiles: application to smoking and lung cancer. *BMC Medical Research Methodology*. 13 (1), 129.

Molitor, J., Brown, I. J., Papathomas, M., Molitor, N., Liverani, S., Chan, Q., Richardson, S., Van Horn, L., Daviglus, M. L., Stamler, J. and Elliott, P. (2014) Blood pressure differences associated with DASH-like lower sodium compared with typical American higher sodium nutrient profile: INTERMAP USA. *Hypertension* 64 (6), 1198-1204. Available at <http://www.ncbi.nlm.nih.gov/pubmed/25201893>

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. *Journal of Statistical Software*, 64(7), 1-30. doi:10.18637/jss.v064.i07.

Hastie, D. I., Liverani, S. and Richardson, S. (2014) Sampling from Dirichlet process mixture models with unknown concentration parameter: Mixing issues in large data implementations. *Statistics and Computing*. Available at <http://link.springer.com/article/10.1007>

Examples

```
## Not run:
# example for Poisson outcome and Discrete covariates
inputs <- generateSampleDataFile(clusSummaryPoissonDiscrete())
runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=10, nClusInit=20,
  nBurn=20, data=inputs$inputData, output="output",
  covNames = inputs$covNames, outcomeT = inputs$outcomeT,
```

```

fixedEffectsNames = inputs$fixedEffectNames)

dissimObj<-calcDissimilarityMatrix(runInfoObj)
clusObj<-calcOptimalClustering(dissimObj)
riskProfileObj<-calcAvgRiskAndProfile(clusObj)
clusterOrderObj<-plotRiskProfile(riskProfileObj,"summary.png")

## End(Not run)

```

calcAvgRiskAndProfile *Calculation of the average risks and profiles*

Description

Calculation of the average risks and profiles.

Usage

```
calcAvgRiskAndProfile(clusObj, includeFixedEffects=F,
  proportionalHazards=F)
```

Arguments

clusObj	Object of type clusObj.
includeFixedEffects	By default this is set to FALSE. If it is set to FALSE then the risk profile is computed with the parameters beta of the fixed effects assumed equal to zero. If it is set to TRUE, then risk profile at each sweep is computed adjusting for the sample of the beta parameter at that sweep.
proportionalHazards	Whether the risk matrix should include lambda only for the yModel="Survival" case so that the proportional hazards can be computed in the plotting function. The default is the average survival time.

Value

A list with the following components. This is an object of type riskProfileObj.

riskProfClusObj	The object of type clusObj as given in the input of this function.
risk	A matrix that has a column for each cluster and a row for each sweep. Each element of the matrix represents the estimated risk at each sweep for each cluster.
profile	An array whose first dimension is the number of sweeps, the second is the number of clusters, the third is the number of discrete covariates and the fourth is the number of categories of each of the covariates. Each element of the array represents the covariate profile at each sweep for each cluster. The fourth dimension does not exists if the covariate type is Normal. If the covariate type is mixed, then instead of this element, the two elements below are defined, 'profilePhi' and 'profileMu'.

profileStar	This is NULL if there has not been any variable selection. otherwise it contains the
empiricals	A vector of length of the optimal number of clusters, where each value is the empirical mean of the outcome for each cluster.
profileStdDev	An array whose first dimension is the number of sweeps, the second is the number of clusters, the third and the fourth are the number of continuous covariates. Each square matrix identified by the first and second dimension of the array represents the standard deviation at each sweep for each cluster. This element is only available if the covariate type is continuous or mixed.
profilePhi	This array is the equivalent of the 'profile' above for discrete covariates in case of mixed covariates.
profileStarPhi	This array is defined as profile and profilePhi, but the values are computed only if a variable selection procedure has been run. The definition of the star profile is given in Liverani, S., Hastie, D. I. and Richardson, S. (2013) PReMiuM: An R package for Bayesian profile regression.
profileMu	This array is the equivalent of the 'profile' above for Normal covariates in case of mixed covariates.
profileStarMu	This array is defined as profile and profileMu, but the values are computed only if a variable selection procedure has been run. The definition of the star profile is given in Liverani, S., Hastie, D. I. and Richardson, S. (2013) PReMiuM: An R package for Bayesian profile regression.
nuArray	For yModel=Survival when weibullFixedShape=FALSE this array contains the sampled values of the shape parameter nu. The first dimension is the number of sweeps, the second is the number of clusters.

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References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. Journal of Statistical Software, 64(7), 1-30. doi:10.18637/jss.v064.i07.

Examples

```
## Not run:
generateDataList <- clusSummaryBernoulliDiscrete()
inputs <- generateSampleDataFile(generateDataList)
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel, nSweeps=10,
  nBurn=20, data=inputs$inputData, output="output", nClusInit=15,
  covNames=inputs$covNames)
```

```

dissimObj<-calcDissimilarityMatrix(runInfoObj)
clusObj<-calcOptimalClustering(dissimObj)
riskProfileObj<-calcAvgRiskAndProfile(clusObj)

## End(Not run)

```

calcDissimilarityMatrix

Calculates the dissimilarity matrix

Description

Calculates the dissimilarity matrix.

Usage

```
calcDissimilarityMatrix(runInfoObj, onlyLS=FALSE)
```

Arguments

runInfoObj	Object of type runInfoObj.
onlyLS	Logical. It is set to FALSE by default. When it is equal to TRUE the dissimilarity matrix is not returned and the only method available to identify the optimal partition using 'calcOptimalClustering' is least squares. This parameter is to be used for datasets with many subjects, as C++ can compute the dissimilarity matrix but it cannot pass it to R for usage in the function 'calcOptimalClustering'. As guidance, be aware that a dataset with 85,000 subjects will require a RAM of about 26Gb, even if onlyLS=TRUE.

Value

Need to write this

disSimRunInfoObj

These are details regarding the run and in the same format as runInfoObj.

disSimMat

The dissimilarity matrix, in vector format. Note that it is diagonal, so this contains the upper triangle diagonal entries.

disSimMatPred

The dissimilarity matrix, again in vector format as above, for the predicted subjects.

lsOptSweep

The optimal partition among those explored by the MCMC, as defined by the least squares method. See Dahl (2006).

onlyLS

Logical. If it set to TRUE the only method available to identify the optimal partition using 'calcOptimalClustering' is least squares.

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Examples

```
## Not run:
generateDataList <- clusSummaryBernoulliDiscrete()
inputs <- generateSampleDataFile(generateDataList)
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel,
  nSweeps=10, nBurn=20, data=inputs$inputData, output="output",
  covNames=inputs$covNames,nClusInit=15)

dissimObj<-calcDissimilarityMatrix(runInfoObj)

## End(Not run)
```

calcOptimalClustering *Calculation of the optimal clustering*

Description

Calculates the optimal clustering.

Usage

```
calcOptimalClustering(disSimObj, maxNClusters=NULL, useLS=F)
```

Arguments

disSimObj	A dissimilarity matrix (in vector format, as the output of the function calcDissimilarityMatrix(), and as described in ?calcDissimilarityMatrix) or a list of dissimilarity matrix, to combine the output of several runs of the MCMC.
maxNClusters	Set the maximum number of clusters allowed. This is set to the maximum number explored.
useLS	This is set to FALSE by default. If it is set to TRUE then the least-squares method is used for the calculation of the optimal clustering, as described in Moliator et al (2010). Note that this is set to TRUE by default if disSimObj\$onlyLS is set to TRUE.

Value

the output is a list with the following elements. This is an object of type `clusObj`.

<code>clusObjRunInfoObj</code>	Details on this run. An object of type <code>runInfoObj</code> .
<code>clusterSizes</code>	Cluster sizes.
<code>clusteringPred</code>	The predicted cluster memberships for the predicted scenarios.
<code>clusObjDisSimMat</code>	Dissimilarity matrix.
<code>clustering</code>	Cluster memberships.
<code>nClusters</code>	Optimal number of clusters.
<code>avgSilhouetteWidth</code>	Average silhouette width when using medoids method for clustering.

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Examples

```
## Not run:
generateDataList <- clusSummaryBernoulliDiscrete()
inputs <- generateSampleDataFile(generateDataList)
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel,
  nSweeps=10, nBurn=20, data=inputs$inputData, output="output",
  covNames=inputs$covNames, nClusInit=15)

dissimObj<-calcDissimilarityMatrix(runInfoObj)
clusObj<-calcOptimalClustering(dissimObj)

## End(Not run)
```

calcPredictions	<i>Calculates the predictions</i>
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Description

Calculates the predictions.

Usage

```
calcPredictions(riskProfObj, predictResponseFileName=NULL,
  doRaoBlackwell=F, fullSweepPredictions=F, fullSweepLogOR=F,
  fullSweepHazardRatio=F, referenceClusterOR=NA)
```

Arguments

riskProfObj	Object of type riskProfObj.
predictResponseFileName	If this function is run after the function profRegr, and outcome (and possibly fixed effects) are known for the predicted profiles, then there is no need to set this, as the function profRegr will have produced a file ending in "_predict-Full.txt". This file allows the computation of measures of fit for cross-validation. If the file has not been produced automatically, it can be produced manually and it can be provided here. We discourage this and we provide no documentation for doing so.
doRaoBlackwell	By default this is set to FALSE. If it is set to TRUE then Rao-Blackwell predictions are computed.
fullSweepPredictions	By default this is set to FALSE. If it is set to TRUE then a prediction is computed for each sweep.
fullSweepLogOR	By default this is set to FALSE. If it is set to TRUE then a prediction log OR is computed for each sweep.
fullSweepHazardRatio	By default this is set to FALSE. If it is set to TRUE then a prediction hazard ratio is computed for each sweep, only for Survival response.
referenceClusterOR	The cluster of reference for the odds ratios. If this is not provided then the first of the predictive profiles provided is used as the reference.

Value

The output is a list with the following elements.

bias	The bias of the predicted values with respect to the observed outcome. If the response is not provided, this is set to NA.
rmse	The root mean square error of the predicted values with respect to the observed outcome. If the response is not provided, this is set to NA.

mae	The mean absolute error of the predicted values with respect to the observed outcome. If the response is not provided, this is set to NA.
observedY	The values of the outcome provided by the user. This is in the case that predictions are run as a validation tool. If the response is not provided, this is set to NA.
predictedY	This matrix has as many rows as predictions requested by the user. It is the median of the predicted values over all the sweeps that have been run after the burn-in period.
doRaoBlackwell	This is set to TRUE if it has done Rao-Blackwell predictions, and FALSE otherwise.
predictedYPerSweep	This array has the first dimension equivalent to the number of sweeps and the second dimension as large as the number of predictions requested by the user. It contains the predicted values per sweep.
logORPerSweep	This array has the first dimension equivalent to the number of sweeps and the second dimension as large as the number of predictions requested by the user. It contains the predicted log OR values per sweep (not available for Poisson and Normal outcome).
fullHR	This array has the first dimension equivalent to the number of sweeps and the second dimension as large as the number of predictions requested by the user. It contains the predicted hazard ratio values per sweep (only for Survival outcome).

Details

This functions computes predicted responses, for various prediction scenarios. It is assumed that the predictive allocations and Rao-Blackwell predictions have already been done in `profRegr` using the 'predict' input.

The user can provide the function `profRegr` with a `data.frame` through the `predict` argument. This `data.frame` has a row for each subject, where each row contains values for the response, fixed effects and offset / number of trials (depending on the response model) where available. Missing values in this `data.frame` are denoted by 'NA'. If the `data.frame` is not provided then the response, fixed effect and offset data is treated as missing for all subjects. If a subject is missing fixed effect values, then the mean value or 0 category fixed effect is used in the predictions (i.e. no fixed effect contribution to predicted response). If the offset / number of trials is missing this value is taken to be 1 when making predictions. If the response is provided for all subjects, the predicted responses are compared with the observed responses and the bias and rmse are computed. If the response is provided in the data frame it must be in a column called "outcome".

The function can produce predicted values based on simple allocations (the default), or a Rao-Blackwellised estimate of predictions, where the probabilities of allocations are used instead of actually performing a random allocation.

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References

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Examples

```
## Not run:
inputs <- generateSampleDataFile(clusSummaryBernoulliDiscrete())

# prediction profiles
preds<-data.frame(matrix(c(0, 0, 1, 0, 0,
0, 0, 1, NA, 0),ncol=5,byrow=TRUE))
colnames(preds)<-names(inputs$inputData)[2:(inputs$nCovariates+1)]

# run profile regression
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel,
  nSweeps=100, nBurn=1000, data=inputs$inputData, output="output",
  covNames=inputs$covNames,predict=preds)

# postprocessing
dissimObj <- calcDissimilarityMatrix(runInfoObj)
clusObj <- calcOptimalClustering(dissimObj)
riskProfileObj <- calcAvgRiskAndProfile(clusObj)
clusterOrderObj <- plotRiskProfile(riskProfileObj,"summary.png",
  whichCovariates=c(1,2))
output_predictions <- calcPredictions(riskProfileObj,fullSweepPredictions=TRUE)

# example where the fixed effects can be provided for prediction
# but the observed response is missing
# (there are 2 fixed effects in this example).
# in this example we also use the Rao Blackwellised predictions

inputs <- generateSampleDataFile(clusSummaryPoissonNormal())

# prediction profiles
predsPoisson<- data.frame(matrix(c(7, 2.27, -0.66, 1.07, 9,
-0.01, -0.18, 0.91, 12, -0.09, -1.76, 1.04, 16, 1.55, 1.20, 0.89,
10, -1.35, 0.79, 0.95),ncol=5,byrow=TRUE))
colnames(predsPoisson)<-names(inputs$inputData)[2:(inputs$nCovariates+1)]

# run profile regression
runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=100,
  nBurn=100, data=inputs$inputData, output="output",
  covNames = inputs$covNames, outcomeT="outcomeT",
  fixedEffectsNames = inputs$fixedEffectNames,predict=predsPoisson)

# postprocessing
```

```

dissimObj<-calcDissimilarityMatrix(runInfoObj)
clusObj<-calcOptimalClustering(dissimObj)
riskProfileObj<-calcAvgRiskAndProfile(clusObj)
output_predictions <- calcPredictions(riskProfileObj,fullSweepPredictions=TRUE)

# example where both the observed response and fixed effects are present
#(there are no fixed effects in this example, but
# these would just be added as columns between the first and last columns).

inputs <- generateSampleDataFile(clusSummaryPoissonNormal())

# prediction profiles
predsPoisson<- data.frame(matrix(c(NA, 2.27, -0.66, 1.07, NA,
    -0.01, -0.18, 0.91, NA, -0.09, -1.76, 1.04, NA, 1.55, 1.20, 0.89,
    NA, -1.35, 0.79, 0.95),ncol=5,byrow=TRUE))
colnames(predsPoisson)<-names(inputs$inputData)[2:(inputs$nCovariates+1)]

# run profile regression
runInfoObj<-profRegr(yModel=inputs$yModel,
    xModel=inputs$xModel, nSweeps=10,
    nBurn=20, data=inputs$inputData, output="output",
    covNames = inputs$covNames, outcomeT="outcomeT",
    fixedEffectsNames = inputs$fixedEffectNames,
    nClusInit=15, predict=predsPoisson)

# postprocessing
dissimObj<-calcDissimilarityMatrix(runInfoObj)
clusObj<-calcOptimalClustering(dissimObj)
riskProfileObj<-calcAvgRiskAndProfile(clusObj)
output_predictions <- calcPredictions(riskProfileObj,fullSweepPredictions=TRUE)

## End(Not run)

```

clusSummaryBernoulliDiscrete

Sample datasets for profile regression

Description

Definition of skeleton of sample datasets for profile regression.

Usage

```

clusSummaryBernoulliDiscrete()
clusSummaryBernoulliNormal
clusSummaryBernoulliDiscreteSmall()
clusSummaryBinomialNormal()
clusSummaryCategoricalDiscrete()

```

```
clusSummaryNormalDiscrete()
clusSummaryNormalNormal()
clusSummaryNormalNormalSpatial()
clusSummaryPoissonDiscrete()
clusSummaryPoissonNormal()
clusSummaryPoissonNormalSpatial()
clusSummaryVarSelectBernoulliDiscrete()
clusSummaryBernoulliMixed()
clusSummaryWeibullDiscrete()
clusSummaryQuantileNormal()
clusSummaryGammaNormal()
```

Value

The output of these function is a list with the following components. These can be used as inputs for profile regression function `profRegr()`.

<code>outcomeType</code>	The outcome type of the dataset.
<code>covariateType</code>	The covariate type of the dataset.
<code>nCovariates</code>	The number of covariates generated.
<code>nCategories</code>	The number of categories of the covariates if the covariates are discrete or mixed.
<code>nFixedEffects</code>	The number of fixed effects.
<code>fixedEffectsCoeffs</code>	The names of the fixed effects.
<code>missingDataProb</code>	The pobability of generating missing data.
<code>nClusters</code>	The number of clusters.
<code>clusterSizes</code>	The number of observations in each cluster.
<code>clusterData</code>	The dataset, including the outcome, the covariates, the fixed effects, the number of trials (if Binomial outcome) and the offset (for Poisson outcome).
<code>covNames</code>	The names of the covariates of the dataset.
<code>nDiscreteCovs</code>	The number of discrete covariates, if the covariate type is mixed.
<code>nContinuousCovs</code>	The number of continuous covariates, if the covariate type is mixed.
<code>outcomeT</code>	The name of the column of the dataset containing the number of trials (if Binomial outcome) or the offset (for Poisson outcome).
<code>includeCAR</code>	A boolean specifying wether a spatial CAR term is included.
<code>TauCAR</code>	The precision for the spatial CAR term.

Details

`clusSummaryBernoulliDiscrete` generates a dataset with Bernoulli outcome and discrete covariates.
`clusSummaryBernoulliNormal` generates a dataset with Bernoulli outcome and Normal covariates.
`clusSummaryBernoulliDiscreteSmall` generates a dataset with Bernoulli outcome and discrete covariates (with smaller cluster sizes).

clusSummaryBinomialNormal generates a dataset with Binomial outcome and discrete covariates.

clusSummaryCategoricalDiscrete generates a dataset with categorical outcome and discrete covariates.

clusSummaryNormalDiscrete generates a dataset with Normal outcome and discrete covariates.

clusSummaryNormalNormal generates a dataset with Normal outcome and Normal covariates.

clusSummaryNormalNormalSpatial generates a dataset with Normal outcome, Normal covariates and a spatial conditional autoregressive term in the log relative risk.

clusSummaryPoissonDiscrete generates a dataset with Poisson outcome and discrete covariates.

clusSummaryPoissonNormal generates a dataset with Poisson outcome and Normal covariates.

clusSummaryPoissonNormalSpatial generates a dataset with Poisson outcome, Normal covariates and a spatial conditional autoregressive term in the log relative risk.

clusSummaryVarSelectBernoulliDiscrete generates a dataset with Bernoulli outcome and discrete covariates, suitable for variable selection as some covariates are not driving the clustering.

clusSummaryBernoulliMixed generates a dataset with Bernoulli outcome and mixed covariates.

clusSummaryWeibullDiscrete generates a dataset with a Weibull outcome and censored observations.

clusSummaryQuantileNormal generates a dataset with a Quantile outcome.

clusSummaryGammaNormal generates a dataset with a Gamma outcome with scale=1.

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References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papatthomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. Journal of Statistical Software, 64(7), 1-30. doi:[10.18637/jss.v064.i07](https://doi.org/10.18637/jss.v064.i07).

Examples

```
names(clusSummaryBernoulliDiscrete())
```

computeRatioOfVariance
<i>computeRatioOfVariance</i>

Description

Computes of the ratio between the variance of the extra variation and the total variance.

Usage

```
computeRatioOfVariance(runInfoObj)
```

Arguments

This function can only be used when the extra variation is included in the response model.

runInfoObj Object of type runInfoObj

Value

For each sweep this function outputs the ratio between the variance of the thetas' and the sum of the variances of the thetas' and the extra variation epsilon as described in Liverani et al. (2013).

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generateSampleDataFile

Generate sample data files for profile regression

Description

Generation of random sample datasets for profile regression.

Usage

```
generateSampleDataFile(clusterSummary, pQuantile=0.05)
```

Arguments

clusterSummary	A vector of strings of the covariate names as by the column names in the data argument.
pQuantile	pQuantile is the quantile parameter of the Asymmetric Laplace Distribution used to generate data to test the model for the quantiles.

Value

The output of this function is a list with the following elements

yModel	The outcome model according to which the data has been generated.
xModel	The covariate model according to which the data has been generated.
inputData	The data.frame that contains the data.
covNames	The names of the covariates.
fixedEffectNames	The names of the fixed effects.
uCAR	The spatial gaussian effect. It is sample into the intrinsic autoregressive model with precision TauCAR under the constraint that the sum of term is null. Only used if includeCAR is TRUE.
TauCAR	The precision of the spatial CAR effect. Only used if includeCAR is TRUE.
Permutation	A vector of size nSubject given the cluster name of each subject. When spatial CAR is added to the model, for preventing potential identifiability problems, the clusters are randomly distributed within the all subjects. Only used if includeCAR is TRUE.

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Examples

```
# generation of data for clustering

generateDataList <- clusSummaryBernoulliDiscrete()
inputs <- generateSampleDataFile(generateDataList)
```

globalParsTrace	<i>Plot of the trace of some of the global parameters</i>
-----------------	---

Description

Function to plot the trace of some global parameters

Usage

```
globalParsTrace(runInfoObj, parameters = "nClusters", plotBurnIn=FALSE, whichBeta=1)
```

Arguments

This function allows to visualise the trace of the global parameters.

Note that this function has not been optimised for large datasets.

runInfoObj	An object of class runInfoObj.
parameters	The parameter whose trace will be plotted. This can be set equal to "nClusters" (default), "alpha", "mpp" and "beta", as by the model. As beta can be a vector, we advise to also set the option "whichBeta" below to select which fixed effect parameter to visualise in the plot. "mpp" stands for marginal partition posterior, also referred to as marginal model posterior.
plotBurnIn	Set to FALSE (default) it does not plot the trace for the burn in period. Set to TRUE it plots the trace including the burn in period.
whichBeta	Integer which selects which fixed effect parameter is plotted.

Value

Plot of trace of some global parameters.

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Examples

```
## Not run:
# generate simulated dataset
generateDataList <- clusSummaryBernoulliDiscreteSmall()
inputs <- generateSampleDataFile(generateDataList)

# run profile regression
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel,
  nSweeps=10, nBurn=20, data=inputs$inputData, output="output", nFilter=3,
  covNames=inputs$covNames, nClusInit=15, reportBurnIn=FALSE,
  fixedEffectsNames = inputs$fixedEffectNames)

# plot trace for alpha
globalParsTrace(runInfoObj, parameters="alpha", plotBurnIn=FALSE)

## End(Not run)
```

heatDissMat

Plot the heatmap of the dissimilarity matrix

Description

Function to plot the heatmap of the dissimilarity matrix

Usage

```
heatDissMat(dissimObj, main=NULL, xlab=NULL, ylab=NULL)
```

Arguments

dissimObj	An object of class dissimObj.
main	The usual plot option, to be passed to the heatmap function.
ylab	The usual plot option, to be passed to the heatmap function.
xlab	The usual plot option, to be passed to the heatmap function.

Value

Plot of the heatmap of the dissimilarity matrix. This function uses the function 'heatmap' of package 'stats'. Note that this function has not been optimised for large datasets.

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Examples

```
## Not run:
# generate simulated dataset
generateDataList <- clusSummaryBernoulliDiscreteSmall()
inputs <- generateSampleDataFile(generateDataList)

# run profile regression
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel,
  nSweeps=10, nBurn=2000, data=inputs$inputData, output="output",
  covNames=inputs$covNames,nClusInit=15)

# compute dissimilarity matrix
dissimObj<-calcDissimilarityMatrix(runInfoObj)

# plot heatmap
heatDissMat(dissimObj)

## End(Not run)
```

is.wholenumber

Function to check if a number is a whole number

Description

Function to check if a number is whole, accounting for a rounding error.

Usage

```
is.wholenumber(x, tol = .Machine$double.eps^0.5)
```

Arguments

x	The number to be checked.
tol	Tolerance level.

Value

The default method for 'is.wholenumber' returns 'TRUE' if the number provided is a whole number.

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References

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Examples

```
is.wholenumber(4) # TRUE
is.wholenumber(3.4) # FALSE
```

mapforGeneratedData	<i>Map generated data</i>
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Description

Function to draw the map of a vector when data are generated.

Usage

```
mapforGeneratedData(u, del=NULL, palette='RGB', main='' )
```

Arguments

u	A vector of size nSubject to map. The function is only useful when data are generated by generateSampleDataFile.
del	A numeric vector of increasing order given the breaks to color the map. By default the centiles of u are used.
palette	Color palette to be used. Either 'RGB' (default) Red-Green-Blue, or 'BW' for black and white.
main	A string for title.

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Examples

```
## Not run:
inputs=generateSampleDataFile(clusSummaryPoissonNormalSpatial())
mapforGeneratedData(inputs$uCAR)

## End(Not run)
```

margModelPosterior	<i>Marginal Model Posterior</i>
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Description

Compute the marginal model posterior.

Usage

```
margModelPosterior(runInfoObj,allocation)
```

Arguments

runInfoObj	An object of type runInfoObj.
allocation	By default, if allocation is not provided, the _z.txt file is read to compute the marginal model posterior for all the partitions available there. If allocation is equal to a vector that corresponds to a partition, the marginal model posterior is computed for that given partition.

Value

It returns a file in the output folder, with name ending in "_margModPost.txt", that contains the marginal model posterior. It also returns a list. The first argument is called margModPost and it is the mean of the values of the marginal model posterior as they appear in the file ending in "_margModPost.txt" in the output folder. The second argument is an updated runInfoObj which also include some hyperparameter values.

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Examples

```
## Not run:
inputs <- generateSampleDataFile(clusSummaryBernoulliDiscrete())

runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=5,
  nBurn=10, data=inputs$inputData, output="output",
  covNames = inputs$covNames, nClusInit=15,
  fixedEffectsNames = inputs$fixedEffectNames)

margModelPost<-margModelPosterior(runInfoObj)

## End(Not run)
```

plotPredictions	<i>Plot the conditional density using the predicted scenarios</i>
-----------------	---

Description

Plots the conditional density for the predicted scenarios provided. It produces a pdf with a page for each predictive scenario provided. Each page has a plot of the predicted response, in the order as they were provided to the function. Note that fixed effects are not processed in this function. This function has been developed for Bernoulli, Normal and Survival response only. This function has been developed for Discrete and Normal covariates only.

Usage

```
plotPredictions(outfile, runInfoObj, predictions,
  logOR=FALSE)
```

Arguments

outfile	String. The name of the output PDF file. The default is "condDensity.pdf".
runInfoObj	An object of type runInfoObj which contains all the details about the run of profRegr.
predictions	An object of type predictions which contains all the details about the run of calcPredictions.
logOR	Whether to plot the response probability or log odds ratios. The default is FALSE and the response probability is plotted.

Value

The output is a plot in PDF format.

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Examples

```
## Not run:
# example with Bernoulli outcome and Discrete covariates
inputs <- generateSampleDataFile(clusSummaryBernoulliDiscrete())
# prediction profiles
preds<-data.frame(matrix(c(
  2, 2, 2, 2, 2,
  0, 0, NA, 0, 0),ncol=5,byrow=TRUE))

colnames(preds)<-names(inputs$inputData)[2:(inputs$nCovariates+1)]
# run profile regression
runInfoObj<-profRegr(yModel=inputs$yModel, xModel=inputs$xModel,
  nSweeps=10000, nBurn=10000, data=inputs$inputData, output="output",
  covNames=inputs$covNames,predict=preds,
  fixedEffectsNames = inputs$fixedEffectNames)
dissimObj <- calcDissimilarityMatrix(runInfoObj)
clusObj <- calcOptimalClustering(dissimObj)
riskProfileObj <- calcAvgRiskAndProfile(clusObj)
predictions <- calcPredictions(riskProfileObj,fullSweepPredictions=TRUE,fullSweepLogOR=TRUE)

plotPredictions(outfile="predictiveDensity.pdf",runInfoObj=runInfoObj,
  predictions=predictions,logOR=TRUE)

## End(Not run)
```

plotRiskProfile

Plot the Risk Profiles

Description

Plots the risk profiles for a profile regression model.

Usage

```
plotRiskProfile(riskProfObj, outFile, showRelativeRisk=F,
               orderBy=NULL, whichClusters=NULL,
               whichCovariates=NULL, useProfileStar=F, riskLim=NULL)
```

Arguments

<code>riskProfObj</code>	An object of type <code>riskProfObj</code> .
<code>outFile</code>	Path and file name to save the plot.
<code>showRelativeRisk</code>	Whether to show the relative risk (with respect to the risk of the first cluster). This option is not available for Normal outcome. For Survival outcomes it computed proportional hazards, but only if the option <code>proportionalHazards=T</code> was used in the function <code>calcAvgRiskAndProfile()</code> .
<code>orderBy</code>	Order by which the clusters are to be displayed. It can take values "Empirical", "ClusterSize" and "Risk" (the latter only if the outcome is provided). It can also take the name of a covariate to order the clusters, in which case the clusters are ordered.
<code>whichClusters</code>	Either a vector of indeces that corresponds to the clusters that are to be displayed. The length of this vector must be greater than 1. The default is that all clusters are shown.
<code>whichCovariates</code>	Either a vector of indeces or a vector of strings that corresponds to the covariates that are to be displayed. The length of this vector must be greater than 1. The default is that all covariates are shown.
<code>useProfileStar</code>	To be set equal to TRUE only if a variable selection procedure has been run. The definition of the star profile is given in Liverani, S., Hastie, D. I. and Richardson, S. (2013) PReMiuM: An R package for Bayesian profile regression.
<code>riskLim</code>	Limits of the y-axis for the plot of the boxplots for the response variable. The default is NULL. If the <code>riskLim</code> are provided, they should be a vector of length 2.

Value

This function creates a png plot saved in the path given by `outFile`. All clusters are visually displayed together.

For discrete covariates, instead of plotting the probability that a ϕ is above or below the mean value, we plot the actual ϕ values (and plot the mean value across clusters as a horizontal line).

For normal covariates, for each covariate the upper plot is the posterior distribution for the mean μ , and the lower plot is the posterior distribution of $\sqrt{\text{Sigma}[j,j]}$ (i.e. the standard deviation for that covariate).

The coloured points on the boxplots highlight the 5

It also returns the following vector.

<code>meanSortIndex</code>	This vector is the index that represents the order that the clusters are represented. The default ordering is by empirical risk.
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Examples

```
## Not run:
# example for Poisson outcome and Discrete covariates
inputs <- generateSampleDataFile(clusSummaryPoissonDiscrete())
runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=10, nClusInit=15,
  nBurn=20, data=inputs$inputData, output="output",
  covNames = inputs$covNames, outcomeT = inputs$outcomeT,
  fixedEffectsNames = inputs$fixedEffectNames)

dissimObj<-calcDissimilarityMatrix(runInfoObj)
clusObj<-calcOptimalClustering(dissimObj)
riskProfileObj<-calcAvgRiskAndProfile(clusObj)
clusterOrderObj<-plotRiskProfile(riskProfileObj,"summary.png")

## End(Not run)
```

profRegr

Profile Regression

Description

Fit a profile regression model.

Usage

```
profRegr(covNames, fixedEffectsNames, outcome="outcome",
  outcomeT=NA, data, output="output", hyper, predict,
  predictType="RaoBlackwell",
  nSweeps=1000, nBurn=1000, nProgress=500, nFilter=1,
  nClusInit, seed, yModel="Bernoulli", xModel="Discrete",
  sampler="SliceDependent", alpha=-2, dPitmanYor = 0, excludeY=FALSE,
  extraYVar=FALSE, varSelectType="None", entropy, reportBurnIn=FALSE,
  run=TRUE, discreteCovs, continuousCovs, whichLabelSwitch="123",
```

```
includeCAR=FALSE, neighboursFile="Neighbours.txt", uCARinit=FALSE,
PoissonCARadaptive=FALSE, weibullFixedShape=TRUE,
useNormInvWishPrior=FALSE, useHyperpriorR1=TRUE,
useIndependentNormal=FALSE, useSeparationPrior=FALSE)
```

Arguments

covNames	A vector of strings of the covariate names as by the column names in the data argument. The names of the covariates cannot include space characters.
fixedEffectsNames	A vector of strings of the fixed effect names as by the column names in the data argument. Each fixed effect must be of class 'numeric'. If a fixed effect is of class 'character', an error message will appear and the fixed effect will need to be recoded as numeric. The names of the fixed effects cannot include space characters.
outcome	A string of column of the data argument that contains the outcome. The outcome cannot have missing values - you could consider predicting the value of the outcome for those subjects for which it has not been observed. The name cannot include space characters.
outcomeT	A string of column of the data argument that contains the offset (for Poisson outcome) or the number of trials (for Binomial outcome) or censoring for Survival response (coded as 0 or 1). The name cannot include space characters.
data	A data frame which has as columns the outcome, the covariates, the fixed effects if any and the offset (for Poisson outcome) or the number of trials (for Binomial outcome) or censoring (for Survival outcome). The outcome cannot have missing values - you could consider predicting the value of the outcome for those subjects for which it has not been observed. For Survival response censoring must be coded as 0 if the event has not occurred (ie, there has been censoring) and 1 if the event has occurred (no censoring has taken place). The names of the columns cannot include space characters.
output	Path to folder to save all output files. The covariates can have missing values, which must be coded as 'NA'. There cannot be missing values in the fixed effects - if there are, use an imputation method before using profile regression.
hyper	Object of type setHyperparams with hyperparameters specifications. This is optional, default values are provided for all hyperparameters. See ?setHyperparams for details.
predict	Data frame containing the predictive scenarios. This is only required if predictions are requested. At each iteration the predictive subjects are assigned to one of the current clusters according to their covariate profiles (but ignoring missing values), or their Rao Blackwellised estimate of theta is recorded (a weighted average of all theta, weighted by the probability of allocation into each cluster. For Normal and Quantile response they can also be randomly allocated. See also the option predictType below. The predictive subjects have no impact on the likelihood and so do not determine the clustering or parameters at each iteration. The predictive allocations are then

recorded as extra entries in each row of the output_z.txt file. This can then be processed in the post processing to create a dissimilarity matrix with the fitting subjects. The post processing function calcPredictions will create predicted response values for these subjects.

See ?calcPredictions for more details and examples.

predictType	This can be set equal to "RaoBlackwell" and "random". The default is RaoBlackwell. The random option can only be used for Normal and Quantile response, where the estimated variance of the clusters is considered and the predictive subjects are randomly assigned to a mixture component and then are also randomly sampled within that component.
nSweeps	Number of iterations of the MCMC after the burn-in period. By default this is 1000.
nBurn	Number of initial iterations of the MCMC to be discarded. By default this is 1000.
reportBurnIn	If TRUE then the burn in iterations are reported in the output files, if set to FALSE they are not. It is set to FALSE by default.
nProgress	The number of sweeps at which to print a progress update. By default this is 500.
nFilter	The frequency (in sweeps) with which to write the output to file. The default value is 1.
nClusInit	The number of clusters individuals should be initially randomly assigned to (Unif[50,60]).
seed	The value for the seed for the random number generator. The default value is the current time.
yModel	The model type for the outcome variable. The options currently available are "Bernoulli", "Poisson", "Binomial", "Categorical", "Normal", "Quantile" and "Survival". The default value is Bernoulli.
xModel	The model type for the covariates. The options currently available are "Discrete", "Normal" and "Mixed". The default value is "Discrete".
sampler	The sampler type to be used. Options are "SliceDependent", "SliceIndependent" and "Truncated". The default value is "SliceDependent".
alpha	The value to be used if alpha is fixed. If a value smaller than or equal to -1 is used then alpha is random, if dPitmanYor is equal to zero (the random alpha option is available for Dirichlet process prior only). The default value is -2 (random alpha). For fixed alpha, if dPitmanYor is in the interval (0,1) then a Pitman-Yor process prior is used instead of a Dirichlet process prior.
dPitmanYor	The discount parameter for the Pitman-Yor process prior. The default value is 0, which is equivalent to a Dirichlet process prior. This parameter must belong to the interval [0,1) and it must be provided together with a non-negative value for alpha. The Pitman-Yor process prior is only available for non-random parameters. Note that the third label switching move is only available for Dirichlet process priors, so it will not be run if dPitmanYor>0. Therefore setting dPitmanYor to a value greater than zero will force whichLabelSwitch=12.
excludeY	If TRUE only the covariate data X is modelled. By default this is set to FALSE.

extraYVar	If set equal to TRUE extra Gaussian variance is included in the response model. This option is available only for Bernoulli, Binomial and Poisson response. By default the extra Gaussian variance is not included, so extraYVar=FALSE.
varSelectType	The type of variable selection to be used "None", "BinaryCluster" or "Continuous". The "Continuous" variable selection is the implementation of the novel variable selection formulation proposed by Papathomas, Molitor, Hoggart, Hastie, Richardson (2012) "Exploring data from genetic association studies using Bayesian variable selection and the Dirichlet process: application to searching for gene x gene patterns" in Genetic Epidemiology. The "BinaryCluster" variable selection is based on the method proposed by Chung and Dunson (2009) "Nonparametric Bayes conditional distribution modelling with variable selection" in the Journal of the American Statistical Association. Both types of variable selection can be used with discrete, continuous or mixed covariates. The default value is "None".
entropy	If included then we compute allocation entropy. By default the allocation entropy is not included.
run	Logical. If TRUE then the MCMC is run. Set run=FALSE if the MCMC has been run already and it is only required to collect information about the run.
discreteCovs	The names of the discrete covariates among the covariate names, if xModel="Mixed". This and continuousCovs must be defined if xModel="Mixed", while covNames is ignored.
continuousCovs	The names of the discrete covariates among the covariate names, if xModel="Mixed". This and continuousCovs must be defined if xModel="Mixed", while covNames is ignored.
whichLabelSwitch	The label switching moves to run. The options available are moves 1, 2 and 3 ("123"), moves 1 and 2 ("12") and move 3 only ("3"). The moves are described in Hastie et al. (2013). Note that the third label switching move is only available for Dirichlet process priors, so it will not be run if dPitmanYor>0. Therefore setting dPitmanYor to a value greater than zero will force whichLabelSwitch=12.
includeCAR	A boolean specifying whether a conditional autoregressive term should be introduced within the model, to take into account possible spatial correlation within residuals. Only for Poisson and Normal response models.
neighboursFile	The file name of the file specifying neighbourhood graph. It should have the same structure than neighbourhood graph files used in the "INLA" package, and can be produced from a nb object of package "spdep", by the function "nb2INLA" of package "spdep". See ?nb2INLA for details. Each file must have at least one neighbour.
uCARinit	This parameter gives the possibility of giving initialisation values for the spatial residuals u of the spatial CAR. It is set to FALSE by default (meaning that the spatial residuals are initialised randomly). It can be set alternatively to a vector of values, one for each of the observations available.
PoissonCARadaptive	This parameter controls which sampler is used for the parameters of the spatial random effect when the outcome is Poisson. When it is set to TRUE, the adaptive rejection sampler is used. When it is set to FALSE (default) a random walk Metropolis is used.

<code>weibullFixedShape</code>	This parameter controls whether the shape parameter of the Weibull distribution (for <code>yModel=Survival</code> only) is a global parameter (fixed) or cluster specific. It is equal to <code>TRUE</code> by default.
<code>useNormInvWishPrior</code>	By default this variable equals <code>FALSE</code> . When this variable equals <code>TRUE</code> , the conjugate Normal-inverse-Wishart prior is used rather than the independent normal and inverse Wishart priors. If this prior is used, variable selection cannot be used as it has not been implemented.
<code>useHyperpriorR1</code>	Adds hyperpriors for the hyperparameter <code>R1</code> , <code>kappa1</code> , <code>mu0</code> and <code>Sigma0</code> for <code>xModel=Normal</code> or <code>Mixed</code> . The default for this option is <code>TRUE</code> .
<code>useIndependentNormal</code>	If the data contains continuous variables (<code>xModel=Normal</code> or <code>Mixed</code>) and the variables are assumed to be independent for each cluster, the multivariate normal likelihood should be replaced by the independent normal likelihood. Therefore, this option should set to <code>TRUE</code> . The default for this option is <code>FALSE</code> . When <code>useIndependentNormal=TRUE</code> , <code>useHyperpriorR1</code> must be <code>TRUE</code> .
<code>useSeparationPrior</code>	A separation prior is used to model the within-cluster covariance matrix for each cluster when the data contains continuous variables (<code>xModel=Normal</code> or <code>Mixed</code>). The default for this option is <code>FALSE</code> . When <code>useSeparationPrior=TRUE</code> , <code>useHyperpriorR1</code> must be <code>TRUE</code> .

Value

Once the C++ has completed the output from fitting the regression is stored in a number of text files in the directory specified. Files are produced containing the MCMC traces for all of the values of interest, along with a log file and files for monitoring the acceptance rates of the adaptive Metropolis Hastings moves.

It returns a number of files in the output directory as well as a list with the following elements. This an object of type `runInfoObj`. The files that are produced in the output directory are described below.

<code>directoryPath</code>	String. Directory path of the output files.
<code>fileStem</code>	String. The
<code>inputFileName</code>	String. Location and file name of input dataset as created by this function for the C++ routines
<code>nSweeps</code>	Integer. The number of sweeps of the MCMC after the burn-in.
<code>nBurn</code>	Integer. The number of iterations in the burn-in period of the MCMC.
<code>reportBurnIn</code>	Logical. Whether the output of the burn-in report should be included.
<code>nFilter</code>	Integer. The frequency (in sweeps) with which to write the output to file.
<code>nProgress</code>	The number of sweeps at which to print a progress update.
<code>nSubjects</code>	Integer. The number of subjects.
<code>nPredictSubjects</code>	Integer. The number of subjects for which to run predictions.

fullPredictFile	Logical. It is FALSE by default. It is equal to TRUE if the outcome or the outcome and the fixed effects were included in the dataframe provided in the input predict. If TRUE, the function will have produced a file ending in "_predictFull.txt" which contains the values of the outcome and fixed effects for the computation of measures of fit in the function calcPredictions.
covNames	A vector of strings with the names of the covariates.
xModel	String. The model type for the covariates.
includeResponse	Logical. If FALSE only the covariate data X is modelled.
yModel	String. The model type for the outcome.
varSelect	Logical. If FALSE no variable selection is performed.
varSelectType	String. It specifies what type of variable selection has been performed, if any.
nCovariates	Integer. The number of covariates.
nFixedEffects	Integer. The number of fixed effects.
nCategoriesY	Integer. The number of categories of the outcome, if yModel = "Categorical". It is 1 otherwise.
nCategories	Vector of integers. The number of categories of each covariate, if xModel = "Discrete". It is 1 otherwise.
extraYVar	TRUE if extra Gaussian variance is included in the response model.
xMat	A matrix of the covariate data.
yMat	A matrix of the outcome data, including the offset if the outcome is Poisson, the number of trials if the outcome is Binomial and 0 or 1 for Survival outcome (1 for censored individuals, 0 otherwise).
wMat	A matrix of the fixed effect data.
whichLabelSwitch	The label switching moves that have been run. The options available are moves 1, 2 and 3 ("123"), moves 1 and 2 ("12") and move 3 only ("3"). The moves are described in Hastie et al. (2013).
includeCAR	Logical. Whether a spatial CAR term is included.
predictType	String. Whether a RaoBlackwell or random predictions have been computed.
weibullFixedShape	Logical. Whether the shape parameter of the Weibull distribution for the survival response is fixed or cluster specific.

These are the files produced in the output directory. We refer to Liverani et al. (2015)

_alpha.txt	If alpha is random, each row is a draw from a posterior distribution of alpha (including burn in if reportBurnIn=TRUE).
_beta.txt	If fixed effects are included, this file provides the draws from the posterior distribution of the beta parameters at each sweep. Each row represents the vector of beta's at each sweep (including burn in if reportBurnIn=TRUE).
_hyper.txt	Internal file to communicate between R and C++ the values of the hyperparamters.

<code>_input.txt</code>	Internal file to communicate the data between R and C++.
<code>_log.txt</code>	This file logs some information about the run, such as what variables were included, which hyperparameters were used, the seed of the random numbers, the acceptance rates of the MCMC moves that were included in the run.
<code>_logPost.txt</code>	This file report the logPosterior, the logLikelihood and logPrior for the model fit at each sweep (including burn in if <code>reportBurnIn=TRUE</code>).
<code>_nClusters.txt</code>	This file includes the number of clusters at each sweep. Each row represents a sweep (including burn in if <code>reportBurnIn=TRUE</code>) and each element in the rows is the number of clusters per sweep. This includes the number of empty clusters, if any.
<code>_nMembers.txt</code>	This file includes the number of observations in each cluster at each sweep. Each row represents a sweep (including burn in if <code>reportBurnIn=TRUE</code>) and each element in the rows is the number of observations in each cluster per sweep. The last number in each row is the total number of observations, computed as the sum of the elements in the row as a check that all observations have been assigned to a cluster.
<code>_theta.txt</code>	This file includes the value of theta (cluster specific parameter for the response variable) for each cluster at each sweep. Each row represents a sweep (including burn in if <code>reportBurnIn=TRUE</code>) and each element in the rows is the value of theta for each cluster at that sweep. The thetas provided here are in the same order as the clusters in <code>_nMembers.txt</code> and they are drawn from the prior when they correspond to empty clusters.
<code>_z.txt</code>	This file includes the cluster membership for each observation at each sweep. Each row represents a sweep (including burn in if <code>reportBurnIn=TRUE</code>) and each element in the rows is the cluster membership for each of the observations, ordered as they are provided to <code>profRegr</code> in the dataframe.

There are more files that can be in the output, depending on which options are used in `profRegr`. The file `_mu.txt` for example reports the mean for `xModel=Normal`, `_phi.txt` reports the multinomial probabilities for `xModel=Discrete`, `_rho.txt` reports the parameters for variable selection, etc. The files usually report one line for each sweep (including burn in if `reportBurnIn=TRUE`). See Liverani et al. (2015) for more details of the parameters.

Note that for the `_gamma.txt` for variable selection the results are reported per sweep (each line is a sweep) and within each line by cluster (so for each covariate the switches per cluster are reported in order, before the second covariate is reported for each cluster, etc).

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The R package PReMiuM is supported through research grants. One key requirement of such funding applications is the ability to demonstrate the impact of the work we seek funding for can. Whatever you are using PReMiuM for, it would be very helpful for us to learn about our users, to tailor our future methodological developments to your needs. Please email us at liveranis@gmail.com or visit <http://www.silvialiverani.com/support-premium/>.

References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. *Journal of Statistical Software*, 64(7), 1-30. doi:10.18637/jss.v064.i07.

Hastie, D. I., Liverani, S. and Richardson, S. (2014) Sampling from Dirichlet process mixture models with unknown concentration parameter: Mixing issues in large data implementations. *Statistics and Computing*. Available at <http://link.springer.com/article/10.1007>

Examples

```
## Not run:
# example for Poisson outcome and Discrete covariates
inputs <- generateSampleDataFile(clusSummaryPoissonDiscrete())
runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=10, nClusInit=20,
  nBurn=20, data=inputs$inputData, output="output",
  covNames = inputs$covNames, outcomeT = inputs$outcomeT,
  fixedEffectsNames = inputs$fixedEffectNames)

# example with Bernoulli outcome and Mixed covariates
inputs <- generateSampleDataFile(clusSummaryBernoulliMixed())
runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=10, nClusInit=15,
  nBurn=20, data=inputs$inputData, output="output",
  discreteCovs = inputs$discreteCovs,
  continuousCovs = inputs$continuousCovs)

## End(Not run)
```

rALD

Asymmetric Laplace Distribution

Description

Random generation and quantile function for a Three Parameter Asymmetric Laplace Distribution as defined in Koenker and Machado (1999) for quantile regression with location parameter equal to μ , scale parameter σ and skewness parameter p .

Usage

```
rALD(n, mu = 0, sigma = 1, p = 0.5)
qALD(prob, mu = 0, sigma = 1, p = 0.5, lower.tail = TRUE)
```

Arguments

<code>prob</code>	Vector of probabilities.
<code>n</code>	Number of observations.
<code>mu</code>	Location parameter.
<code>sigma</code>	Scale parameter.
<code>p</code>	Skewness parameter.
<code>lower.tail</code>	Logical; if TRUE (default), probabilities are $P[X \text{ strictly smaller than } x]$ otherwise, $P[X > x]$.

Details

If `mu`, `sigma` or `p` are not specified they assume the default values of 0, 1 and 0.5, respectively, belonging to the Symmetric Standard Laplace Distribution denoted by $ALD(0,1,0.5)$.

As discussed in Koenker and Machado (1999) and Yu and Moyeed (2001) we say that a random variable Y is distributed as an ALD with location parameter μ , scale parameter $\sigma > 0$ and skewness parameter p in $(0,1)$, if its probability density function (pdf) is given by

$$f(y|\mu, \sigma, p) = p(1-p)/\sigma * e^{-(p_p(y-\mu)/\sigma)}$$

where $p_p(\cdot)$ is the so called check (or loss) function defined by

$$p_p(u) = u(p - I(u < 0))$$

with $I(\cdot)$ denoting the usual indicator function. This distribution is denoted by $ALD(\mu, \sigma, p)$ and its p -th quantile is equal to μ .

The scale parameter σ must be positive and non zero. The skew parameter p must be between zero and one ($0 < p < 1$).

Value

The length of the result is determined by `n` for `rALD`, and is the maximum of the lengths of the numerical arguments for the other functions `dALD`, `pALD` and `qALD`.

Authors

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References

Galarza Morales, C., Lachos Davila, V., Barbosa Cabral, C., and Castro Cepero, L. (2017) Robust quantile regression using a generalized class of skewed distributions. Stat,6: 113-130 doi:[10.1002/sta4.140](https://doi.org/10.1002/sta4.140).

Examples

```
is.wholenumber(4) # TRUE
is.wholenumber(3.4) # FALSE
```

setHyperparams

*Definition of characteristics of sample datasets for profile regression***Description**

Hyperparameters for the priors can be specified here and passed as an argument to profRegr.

The user can specify some or all hyperparameters. Those hyperparameters not specified will take their default values. Where the file is not provided, all hyperparameters will take their default values.

Usage

```
setHyperparams(shapeAlpha=NULL,rateAlpha=NULL,
  aPhi=NULL,mu0=NULL,Tau0=NULL, TauIndep0 = NULL, R0=NULL,
  RIndep0 = NULL, kappa0=NULL, kappa1=NULL,
  nu0=NULL,muTheta=NULL,sigmaTheta=NULL,dofTheta=NULL,muBeta=NULL,
  sigmaBeta=NULL,dofBeta=NULL,shapeTauEpsilon=NULL,
  rateTauEpsilon=NULL,aRho=NULL,bRho=NULL,atomRho=NULL,shapeSigmaSqY=NULL,
  scaleSigmaSqY=NULL,pQuantile=NULL,rSlice=NULL,truncationEps=NULL,
  shapeTauCAR=NULL,rateTauCAR=NULL,shapeNu=NULL,scaleNu=NULL,
  initAlloc=NULL)
```

Arguments

shapeAlpha	The shape parameter for Gamma prior on alpha (default=2)
rateAlpha	The inverse-scale (rate) parameter for the Gamma prior on alpha (default=1)
aPhi	The vector of parameters for the Dirichlet prior on phi_j. Element j corresponds to covariate j which then has a prior Dirichlet(aPhi[j],aPhi[j],...,aPhi[j]). Only used in discrete case, default=(1 1 1 ... 1).
mu0	The mean vector for mu_c in the multivariate Normal covariate case (only used in multivariate Normal covariate case (useIndependentNormal=FALSE), default=empirical covariate means)
Tau0	The precision matrix for mu_c in the multivariate Normal covariate case (only used in multivariate Normal covariate case, when useIndependentNormal=FALSE). The default value is default=inverse of diagonal matrix with elements equal to square of empirical range for each covariate
TauIndep0	The precision parameter of each covariate (in a vector form) for mu_c in the independent Normal covariate case (only used in independent Normal covariate case, when useIndependentNormal=TRUE). The default value is default=a vector with elements equal to inverse of the square of empirical range for each covariate)
R0	The scale parameter for the Wishart distribution for Tau_c if useHyperpriorR1=FALSE in the function profRegr. If useHyperpriorR1=TRUE in the function profRegr, then R0 is the scale parameter for the prior distribution on the scale parameter of the precision matrix Tau_c (in this case Tau_c has Wishart distribution with

	parameters R_0 and κ_0). In both cases the default is $\text{default}=1/n\text{Covariates}$ * inverse of empirical covariance matrix. These parameters can only be used for Normal or Mixed covariates.
RIndep0	The rate parameter in the gamma distribution for R_1 indep of each covariate (in a vector form) if <code>useIndependentNormal=TRUE</code> in the function <code>profRegr</code> . The default is <code>default=</code> a vector with elements equal to $10/\text{square of empirical range}$ for each covariate. The parameter can only be used for Normal or Mixed covariates.
κ_0	The degrees of freedom for the Wishart distribution for Tau_c if <code>useHyperpriorR1=FALSE</code> in the function <code>profRegr</code> . If <code>useHyperpriorR1=TRUE</code> in the function <code>profRegr</code> , then κ_0 are the degrees of freedom for the prior distribution on the scale parameter of the precision matrix Tau_c (in this case Tau_c has Wishart distribution with parameters R_0 and κ_0). In both cases the default is $n\text{Covariates}$. These parameters can only be used for Normal or Mixed covariates.
κ_1	The degrees of freedom parameter for the Wishart distribution for Tau_c (only used in Normal covariate case, <code>default=nCovariates</code>). Only used when the prior for R_1 is included in the model (by setting the option <code>useHyperpriorR1=TRUE</code> in the function <code>profRegr</code>).
ν_0	Hyperparameter for the conjugate Normal inverse Wishart prior for Normal covariates. The Normal distribution of μ_c has covariance Sigma_c/ν_0 . The default value is 0.01. The other hyperparameters for this parametrisation are reused from the independent priors. This hyperparameter is only useful when the option <code>useNormInvWishPrior=TRUE</code> in the function <code>profRegr</code> .
μ_{Theta}	The location parameter for the t-Distribution for θ_c (only used if response included in model, <code>default=0</code>)
σ_{Theta}	The scale parameter for the t-Distribution for θ_c (only used if response included in model, <code>default=2.5</code>)
$\text{dof}_{\text{Theta}}$	The degrees of freedom parameter for the t-Distribution for θ_c (only used if response included in model, <code>default=7</code>)
μ_{Beta}	The location parameter for the t-Distribution for β (only used when fixed effects present, <code>default=0</code>)
σ_{Beta}	The scale parameter for the t-Distribution for β (only used when fixed effects present, <code>default=2.5</code>)
dof_{Beta}	The dof parameter for the t-Distribution for β (only used when fixed effects present, <code>default=7</code>)
shapeTauEpsilon	Shape parameter for gamma distribution for prior for precision tau of extra variation errors epsilon (only used if extra variation is used i.e. <code>extraYVar</code> argument is included, <code>default=5.0</code>)
rateTauEpsilon	Inverse-scale (rate) parameter for gamma distribution for prior for precision tau of extra variation errors epsilon (only used if extra variation is used i.e. <code>extraYVar</code> argument is used, <code>default=0.5</code>)
aRho	Parameter for beta distribution for prior on rho in variable selection (<code>default=0.5</code>)

bRho	Parameter for beta distribution for prior on rho in variable selection (default=0.5)
atomRho	Parameter for the probability for the atom at zero, i.e. the 0.5 probability in w_j distributed Bernoulli(0.5) in the formulation of the sparsity inducing prior (default=0.5). This parameter must be in the interval (0,1], where atomRho=1 corresponds to the case where the prior for rho is a Beta(aRho,bRho).
shapeSigmaSqY	Shape parameter of inverse-gamma prior for σ_Y^2 (only used in the Normal response model, default =2.5)
scaleSigmaSqY	Scale parameter of inverse-gamma prior for σ_Y^2 (only used in the Normal response model, default =2.5)
pQuantile	Quantile for the yModel=Quantile option (default = 0.5)
rSlice	Slice parameter for independent slice sampler such that $x_{i,c} = (1-rSlice)*rSlice^c$ for $c=0,1,2,\dots$ (only used for slice independent sampler i.e. sampler=SliceIndependent, default 0.75).
truncationEps	Parameter for determining the truncation level of the finite Dirichlet process (only used for truncated sampler i.e. sampler=Truncated)
shapeTauCAR	Shape parameter for gamma distribution for precision TauCAR of spatial CAR term (only used if a spatial term is included i.e. includeCAR argument is TRUE, default=0.001)
rateTauCAR	Inverse-scale (rate) parameter for gamma distribution for precision TauCAR of spatial CAR term (only used if a spatial term is included i.e. includeCAR argument is TRUE, default=0.001)
shapeNu	Shape parameter of Gamma prior for the shape parameter of the Weibull for survival response (only used in the Survival response model, default = 2.5)
scaleNu	Scale parameter of Gamma prior for the shape parameter of the Weibull for survival response (only used in the Survival response model, default = 1)
initAlloc	Vector of the initial allocation of the individuals to clusters. This is NULL by default, which implies a random start. Useful for starting the MCMC from a specific partition. Note that if this overwrites the option nClusInit in the function profRegr: nClusInit is set equal to the maximum value in initAlloc.

Value

The output of this function is a list with the components defined as above.

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References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. Journal of Statistical Software, 64(7), 1-30. doi:10.18637/jss.v064.i07.

Examples

```
## Not run:
hyp <- setHyperparams(shapeAlpha=3,rateAlpha=2,mu0=c(30,13),R0=3.2*diag(2))

inputs <- generateSampleDataFile(clusSummaryPoissonNormal())
runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=2, nClusInit=15,
  nBurn=2, data=inputs$inputData, output="output",
  covNames = inputs$covNames, outcomeT = inputs$outcomeT,
  fixedEffectsNames = inputs$fixedEffectNames,
  hyper=hyp)

## End(Not run)
```

simBenchmark

Benchmark for simulated examples

Description

This function checks the cluster allocation of profile regression against the generating clusters for a selection of the simulated dataset provided within the package. This can be used to compute confusion matrices for simulated examples, as shown in the example below.

Usage

```
simBenchmark(whichModel = "clusSummaryBernoulliDiscrete",
  nSweeps = 1000, nBurn = 1000, seedProfRegr = 123)
```

Arguments

whichModel	Which simulated dataset this benchmark should be carried out for. At the moment this function works only for these datasets structures provided in the package: "clusSummaryBernoulliNormal","clusSummaryBernoulliDiscreteSmall","clusSummaryCategorical","clusSummaryNormalDiscrete","clusSummaryNormalNormal", "clusSummaryNormalNormalSpatial", "clusSummaryVarSelectBernoulliDiscrete", "clusSummaryBernoulliMixed". These dataset structures can be used by the function generateSampleDataFile to create simulated datasets.
nSweeps	The number of sweeps of the profile regression algorithm for this benchmarking.
nBurn	The number of sweeps in the burn in of the profile regression algorithm for this benchmarking.
seedProfRegr	Sets the seed for the random number generation in profile regression (ie. sets the seed for the portion of the MCMC code in C++). Note that setting this seed does not mean that the function simBenchmark will give the same answer. This is because the first step of this function generates a random sample, which will vary in each run unless a global seed is set in R using the function set.seed.

Value

This function creates a data.frame. Each row corresponds to each observation in the generated dataset. The columns are:

clusterAllocation	Cluster allocation carried out by profile regression. These values are integers, corresponding to cluster numbers.
outcome	Value of the outcome (y) in the dataset.
generatingCluster	Cluster allocation in the data generating mechanism. These values are characters which include the word 'Known' and then the original numbering of the cluster. The word 'Known' is included to avoid confusion with the cluster allocations identified by profile regression.

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References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. Journal of Statistical Software, 64(7), 1-30. doi:[10.18637/jss.v064.i07](https://doi.org/10.18637/jss.v064.i07).

Examples

```
## Not run:
# vector of all test datasets allowed by this benchmarking function
testDatasets<-c("clusSummaryBernoulliNormal",
  "clusSummaryBernoulliDiscreteSmall", "clusSummaryCategoricalDiscrete",
  "clusSummaryNormalDiscrete", "clusSummaryNormalNormal",
  "clusSummaryNormalNormalSpatial", "clusSummaryVarSelectBernoulliDiscrete",
  "clusSummaryBernoulliMixed")

# runs profile regression on all datasets and
# computes confusion matrix for each one
for (i in 1:length(testDatasets)){
  tester<-simBenchmark(testDatasets[i])
  print(table(tester[,c(1,3)]))
}

## End(Not run)
```

`summariseVarSelectRho` *summariseVarSelectRho*

Description

This function summarises the posterior distribution of rho, a parameter for variable selection only.

Usage

```
summariseVarSelectRho(runInfoObj)
```

Arguments

`runInfoObj` Object of type `runInfoObj`

Value

A list with the following elements.

<code>rho</code>	A matrix that has as many columns as the number of covariates and as many rows as the number of sweeps. This matrix records the samples from the posterior distribution of rho for each covariate at each sweep.
<code>rhoMean</code>	Vector with the column means of the matrix rho above. Each value corresponds to the posterior mean of rho for each covariate.
<code>rhoMedian</code>	Vector with the column medians of the matrix rho above. Each value corresponds to the posterior median of rho for each covariate.
<code>rhoLowerCI</code>	Vector with the column lower confidence intervals of the matrix rho above. Each value corresponds to the lower confidence interval of the posterior distribution of rho for each covariate.
<code>rhoUpperCI</code>	Vector with the column upper confidence intervals of the matrix rho above. Each value corresponds to the upper confidence interval of the posterior distribution of rho for each covariate.

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References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. Journal of Statistical Software, 64(7), 1-30. doi:[10.18637/jss.v064.i07](https://doi.org/10.18637/jss.v064.i07).

Examples

```
## Not run:
inputs <- generateSampleDataFile(clusSummaryVarSelectBernoulliDiscrete())

runInfoObj<-profRegr(yModel=inputs$yModel,
  xModel=inputs$xModel, nSweeps=10, nClusInit=15,
  nBurn=20, data=inputs$inputData, output="output",
  covNames = inputs$covNames, varSelect="Continuous")

rho<-summariseVarSelectRho(runInfoObj)

## End(Not run)
```

vec2mat	<i>Vector to upper triangular matrix</i>
---------	--

Description

Function to convert a vector to an upper triangular matrix. The vector does not include the diagonal values, which are then set equal to 1 in the matrix. The matrix is filled by row.

Usage

```
vec2mat(data = NA, nrow = 1)
```

Arguments

data	The vector to be converted, excluding the diagonal which is set equal to 1.
nrow	The number of rows (and columns) of the resulting matrix.

Value

The symmetric matrix. The matrix is filled by column.

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References

Silvia Liverani, David I. Hastie, Lamiae Azizi, Michail Papathomas, Sylvia Richardson (2015). PRReMiuM: An R Package for Profile Regression Mixture Models Using Dirichlet Processes. Journal of Statistical Software, 64(7), 1-30. doi:[10.18637/jss.v064.i07](https://doi.org/10.18637/jss.v064.i07).

Examples

```
vec2mat(data=c(1,2,3),nrow=3)
```

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